Cambridge Assessment International Education

Cambridge International AS & A Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

6281035904

COMPUTER SCIENCE **9618/31**

Paper 3 Advanced Theory **May/June 2024**

1 hour 30 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].
- No marks will be awarded for using brand names of software packages or hardware.

This document has **12** pages. Any blank pages are indicated.

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1 Real numbers are stored in a computer system using floating-point representation with:

- 10 bits for the mantissa
- 6 bits for the exponent
- two's complement form for both the mantissa and the exponent.

(a) Calculate the denary value of the given normalised floating-point number.

Show your working.

Mantissa										Exponent					
0	1	0	0	1	1	1	1	0	0	0	0	1	0	0	1

Working .....

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Answer .....

[3]

(b) Calculate the normalised floating-point representation of –102.75 in this system.

Show your working.

Mantissa										Exponent					

Working .....

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[3]

2 The TCP / IP protocol suite has four layers:

Transport, Application, Link, Internet

(a) Complete the diagram to show the correct order for these layers.



[2]

(b) Describe the function of the Transport layer.

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..... [2]

(c) Outline **one** protocol that is associated with the Application layer.

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..... [2]

3 **(a)** Explain what is meant by **non-composite** and **composite** data types.

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..... [3]

(b) Write **pseudocode** statements to declare the record data type `FootballClub` to hold data about football clubs in a league, to include:

- name of team
- date team joined the league
- main telephone number
- name of the manager
- number of members
- current position in the league.

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..... [4]

4 **(a)** Describe the sequential method of file **access**.

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..... [2]

(b) Explain how the sequential method of file access is applied to files with serial organisation and to files with sequential organisation.

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..... [3]

5 **(a)** Write this Reverse Polish Notation (RPN) in infix form:

```
5 2 + 9 3 - / 3 *
```

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..... [3]

(b) Write this infix expression in RPN:

```
((7 + 3) - (2 * 8)) / 6
```

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..... [2]

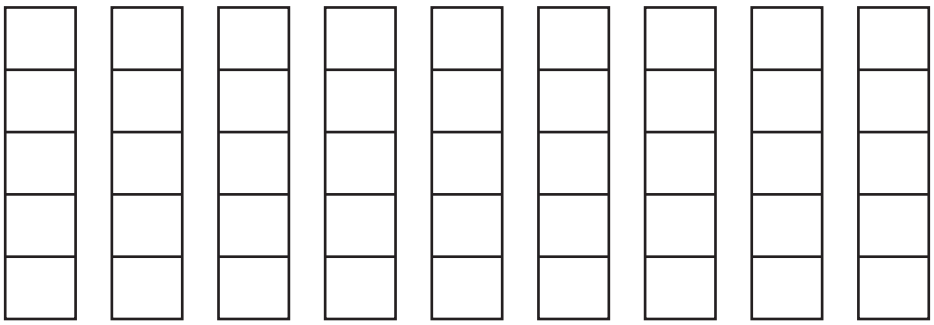
(c) Evaluate this RPN expression:

```
a b - c d + * e /
```

when

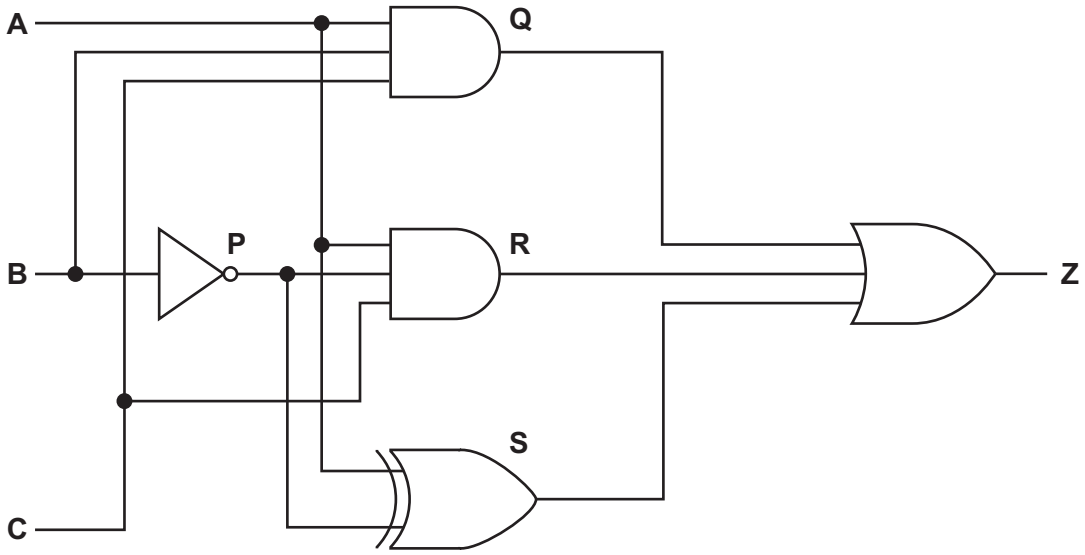
`a = 17, b = 5, c = 7, d = 3` and `e = 10`

Show the changing contents of the stack as the RPN expression is evaluated.



[4]

6 The diagram shows a logic circuit.



(a) Complete the truth table for the given logic circuit.

Show your working.

			Working space				
A	**B**	**C**	**P**	**Q**	**R**	**S**	**Z**
0	0	0					
0	0	1					
0	1	0					
0	1	1					
1	0	0					
1	0	1					
1	1	0					
1	1	1					

[3]

(b) Write the Boolean expression that corresponds to the logic circuit as a sum-of-products.

Z = ...

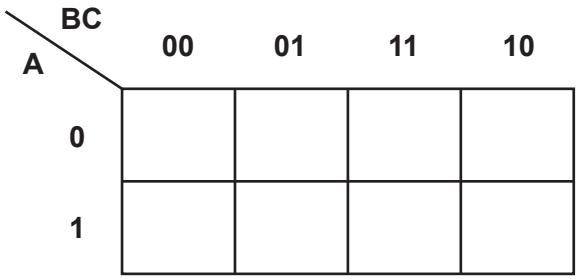
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..... [2]

(c) **(i)** Complete the Karnaugh map (K-map) for the Boolean expression:

$\overline{A}.\overline{B}.\overline{C} + \overline{A}.\overline{B}.C + A.\overline{B}.\overline{C} + A.\overline{B}.C + A.B.\overline{C} + A.B.C$



[2]

(ii) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

(iii) Write the Boolean expression from your answer to part **(c)(ii)** as a simplified sum-of-products.

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..... [1]

7 **(a)** Describe what is meant by a digital certificate.

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..... [3]

(b) Explain the role of a digital certificate in creating a digital signature.

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..... [2]

8 A **declarative** programming language is used to represent the features that are available and the features that are unavailable on different body styles of a car.

```
01 feature(sunroof).
02 feature(automatic_tailgate).
03 feature(heated_seats).
04 feature(extra_seats).
05 feature(reversing_camera).
06 feature(dashboard_camera).
07 feature(air_conditioning).
08 feature(heated_windscreen).
09 feature(satnav).
10 bodystyle(saloon).
11 bodystyle(hatchback).
12 bodystyle(estate).
13 bodystyle(minivan).
14 bodystyle(convertible).
15 available(sunroof, hatchback).
16 available(sunroof, minivan).
17 available(reversing_camera, hatchback).
18 available(extra_seats, minivan).
19 available(reversing_camera, saloon).
20 unavailable(sunroof, convertible).
21 unavailable(automatic_tailgate, saloon).
22 unavailable(extra_seats, hatchback).
```

These clauses have the meanings:

Clause	Meaning
01	Sunroof is a feature.
10	Saloon is a body style.
15	Sunroof is available on a hatchback.
20	Sunroof is unavailable on a convertible.

(a) Sliding doors is a feature that is available on a minivan but unavailable on a hatchback.

Write additional clauses to represent this information.

23 .....

24 .....

25 .....

[3]

(b) Using the variable `Options`, the goal:

```
available(Options, saloon)
```

returns

```
Options = reversing_camera
```

Write the result returned by the goal:

```
available(Options, hatchback)
```

`Options =` ..... [1]

(c) `F` may be available for `B` if `F` is a feature and `B` is a body style and `F` is **not** unavailable for that body style.

Write this as a rule:

```
may_choose_option(F, B)
```

`IF` .....

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..... [4]

9 Explain what is meant by **Deep Learning** in relation to Artificial Intelligence (AI).

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..... [3]

10 **(a)** State a condition that must be true for an array to be searchable for a binary search.

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..... [1]

(b) Complete the given pseudocode to find an item in a 1D array `Names` of type `STRING` using a binary search.

```
DECLARE Names : ARRAY[1:100000] OF STRING
DECLARE TopOfList : INTEGER
DECLARE EndOfList : INTEGER
DECLARE CurrentItem : INTEGER
DECLARE ToFind : STRING
DECLARE Found : BOOLEAN
DECLARE NotInList : BOOLEAN
TopOfList ← 1
EndOfList ← 100000

OUTPUT "Which name do you wish to find? "
INPUT ToFind

.....

NotInList ← FALSE

WHILE ..... AND .....
   CurrentItem ← (TopOfList + EndOfList) DIV 2

   IF ..... THEN
      Found ← TRUE
   ELSE
      IF TopOfList >= EndOfList THEN

         .....
      ELSE
         IF ToFind > Names[CurrentItem] THEN

            .....
         ELSE
            EndOfList ← CurrentItem - 1
         ENDIF
      ENDIF
   ENDIF
ENDWHILE
IF Found = TRUE THEN
   OUTPUT "Item found at position ", CurrentItem, " in array"
ELSE
   OUTPUT "Item not in array"
ENDIF
```

[5]

(c) Describe the performance of a binary search in relation to the number of data items in the array being searched. Refer to Big O notation in your answer.

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..... [2]

11 Reduced Instruction Set Computers (RISC) and Complex Instruction Set Computers (CISC) are two types of processor.

(a) State **two** features of RISC processors.

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..... [2]

(b) Outline the process of interrupt handling as it could be applied to RISC or CISC processors.

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..... [3]

(c) Explain how pipelining affects interrupt handling for RISC processors.

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..... [3]

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